动态随机一般均衡模型：理论与应用
第 9 章：福利分析

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A 动态系统

$$U_t = \ln(C_t) - \frac{N_t^{1+\varphi}}{1+\varphi} + \beta \mathbb{E}_t U_{t+1}$$

$$\beta \mathbb{E}_t \frac{C_t}{C_{t+1}} \frac{1}{\Pi_{t+1}} = \frac{1}{R_t}$$

$$N_t^{\varphi} C_t = \hat{W}_t$$

$$Y_{It} = A_t N_t$$

$$\hat{P}_t^0 = \frac{1-\tau}{\mu} \frac{S_{1t}}{S_{2t}}$$

$$S_{1t} = \frac{Y_t}{C_t} \frac{\hat{W}_t}{A_t} + \beta\theta \mathbb{E}_t \Pi_{t+1}^{\epsilon} S_{1t+1}$$

$$S_{2t} = \frac{Y_t}{C_t} + \beta\theta \mathbb{E}_t \Pi_{t+1}^{\epsilon-1} S_{2t+1}$$

$$\theta \Pi_t^{\epsilon-1} + (1-\theta)(\hat{P}_t^0)^{1-\epsilon} = 1$$

$$Y_t = C_t$$

$$Y_{It} = V_t Y_t$$

$$V_t = (1-\theta)(\hat{P}_t^0)^{-\epsilon} + \theta \Pi_t^{\epsilon} V_{t-1}$$

货币政策

$$\ln\left(\frac{A_t}{A}\right) = \rho \ln\left(\frac{A_{t-1}}{A}\right) + \eta_t^a, \quad \eta_t^a \sim \text{i.i.d.N}(0, \sigma_a^2)$$

B 求解模型稳态

B.1 稳态系统

设稳态时 $A = 1$ 以及 $\Pi = 1$，模型稳态系统为：

$$U = \ln(C) - \frac{N^{1+\varphi}}{1+\varphi} + \beta U$$

$$\beta = \frac{1}{R}$$

$$N^{\varphi} C = \hat{W}_t$$

$$Y_I = AN$$

$$\hat{P}^0 = \frac{1-\tau}{\mu} \frac{S_1}{S_2}$$

$$S_1 = \frac{Y}{C}\hat{W} + \beta\theta S_1$$
$$S_2 = \frac{Y}{C} + \beta\theta S_2$$
$$\theta + (1-\theta)(\hat{P}^0)^{1-\epsilon} = 1$$
$$Y = C$$
$$Y_I = VY$$
$$V = (1-\theta)(\hat{P}^0)^{-\epsilon} + \theta V$$

B.2 稳态均衡

设 $\tau = \frac{1}{\epsilon}$，求得稳态均衡：

$$\hat{P}^0 = V = 1$$
$$\hat{W} = 1$$
$$Y = C = Y_I = N = \hat{W} = 1$$
$$S_1 = S_2 = \frac{1}{1-\beta\theta}$$
$$U = \frac{1}{1-\beta}\left(\ln(C) - \frac{N^{1+\varphi}}{1+\varphi}\right) = -\frac{1}{(1-\beta)(1+\varphi)}$$